MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PULSE MODULATOR

TYPE 3D21WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Beam tetrode, vacuum

Outline --- See figure 1

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	Cap
Element	---	hct	h	nc	g2	nc	g1	h	k	a

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Pg1	Ik	Ic1
Unit:	V	Vdc	Vdc	Vdc	v	W	mAdc	mAdc
Maximum:	13.9 or 6.9	4,000 (see note 1)	0	850	500	0.5	175	5.0
Minimum:	11.4 or 5.7	---	-200 (see note 2)	--- (see note 3)	---	---	---	---
TEST CONDITIONS:	6.3	600	0	300	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Pp	Pg2	egk	ik	tk	TE	Alt
Unit:	W	W	v	a	sec	°C	ft
Maximum:	40	3.5	220	10	---	300	(See note 5)
Minimum:	---	---	---	--- (see figure 2)	30	--- (see note 4)	---
TEST CONDITIONS:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 1							
1266	Total grid current	t = 120; Rk = 825 ohms (see note 6)	0.65	II	Ic1	0	-5.0	μAdc
1256	Electrode current (1) (anode)	Rk = 825 ohms (see note 6)	0.65	II	Ib	28	37	mAdc
Ⓔ 1256	Electrode current (2) (anode)	Ebb = 4,000 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; Rp = 2 Meg (see note 7)	0.65	II	Ib	0	1.0	mAdc
1256	Electrode current (screen)	Rk = 825 ohms	0.65	II	Ic2	---	3.0	mAdc
	Quality conformance inspection, part 2							
---	Pulse operation	Ebb = 4,000 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see notes 7 and 8)	1.0	II	---	---	---	---
Ⓔ 1356	Grid-pulse operation	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see note 8) (test in accordance with figure 4)	1.0	II	ib	5.0	---	a
---	Peak screen current	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see note 8) (test in accordance with figure 4)	1.0	II	ic2	---	4.0	a
---	Peak grid current (1)	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see note 8) (test in accordance with figure 4)	1.0	II	ic1	---	2.0	a
---	Peak grid current (2)	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 52 v (see note 8)	1.0	II	ic1	30	---	ma

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued							
---	Grid current operation	Eb = 250 Vdc; Ec2 = 150 Vdc; Ec1 = -6 Vdc thru a 0.5 Meg resistor; Ef = 6.5 V (see note 9)	---	---	Ic1	---	15	µAdc
1031	Low-frequency vibration	Eb = 250 Vdc; Ec2 = 100 Vdc; Ef = 12.6 V; Ec1 = -10 Vdc; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1301	Heater current		---	---	If	1.55	1.85	A
1306	Transconductance	Rk = 825 ohms	---	---	Sm	5,000	7,000	µmhos
1336	Heater-cathode leakage	Ehk = 450 Vdc	---	---	Ihk	---	50	µAdc
1331	Direct-interelectrode capacitance	No shield	---	---	Cgp	---	1.2	pF
					Cin	13	21	pF
					Cout	7.5	12.5	pF
1041	Shock	450 G; Ehk = 450 Vdc; Ec1 = -10 Vdc; Rp = 2,000 ohms; Eb = 250 Vdc; Ec2 = 100 Vdc	15	S4	---	---	---	---
1031	Vibration-fatigue	5 G; F = 50 to 500 Hz; Ef = 6.3 V; no other voltages applied (see note 11)	6.5	See note 10	---	---	---	---
---	Shock and vibration-fatigue-test end points							
1336	Heater-cathode leakage	Ehk = 450 Vdc	---	---	Ihk	---	100	µAdc
(E) 1356	Grid-pulse operation	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see note 8) (test in accordance with figure 4)	---	---	ib	3.75	---	a
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1216	Base material insulating quality		---	---	---	---	---	---

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 2 -Continued							
1111	Base pin solder depth		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1506	Heater-cycling life	Ef = 7.0 V; Ehk = 300 Vac; Ec2 = Eb = 0; 1 minute "on", 4 minutes "off"	---	---	---	---	---	---
---	Heater-cycling life-test end points							
1336	Heater-cathode leakage	Ehk = 450 Vdc	---	---	Ihk	---	150	µAdc
1301	Heater current		---	---	If	1.4	2.0	A
---	Life test	Group B; pulse operation; Ebb = 3,500 Vdc; Ecc2 = 800 Vdc; eo = 300 v; tp = 4 µs; prr = 250; R1 = 500 ohms ±5% (see note 12)	---	---	---	---	---	---
---	Life-test end point (500 hours)							
(E) 1356	Grid-pulse operation	Ebb = 420 Vdc; Ecc2 = 800 Vdc; Ecc1 = -150 Vdc; egy = 155 v (see note 8) (test in accordance with figure 4)	---	---	ib	3.75	---	a

NOTES:

1. Instantaneous anode voltage due to transient shall not exceed 5,000 volts. Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere. With a screen potential not exceeding 400 Vdc and when no instantaneous voltage due to transient is present (essentially resistance anode load), a maximum anode potential of 4,500 Vdc may be used.

2. Instantaneous negative grid voltage due to transient shall not exceed -500 volts. Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere.

3. Series resistance shall be inserted in the power supply sufficient to limit the short circuit current to less than 0.5 ampere.

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4. Cooling shall be employed if the additive effects of simultaneous operation at high altitude and high dissipation or proximity to other heat sources in the equipment tend to cause the maximum bulb temperature to be exceeded.

5. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

6. This test shall be performed at the conclusion of the holding period.

7. The circuit shown on figure 3, or the equivalent, shall be used for high-voltage anode current (2) test and high voltage pulse operation test.

8. After a minimum preheat of 30 seconds using Ef = 6.3 volts only, the tube shall be tested under the following conditions: A substantially rectangular pulse with a width of 4 µs and a prr of 250 shall be applied to the grid of the tube under test. Initial arcing may be tolerated, but the tube shall be free of such arcing during final 30 seconds of a 1-minute test.

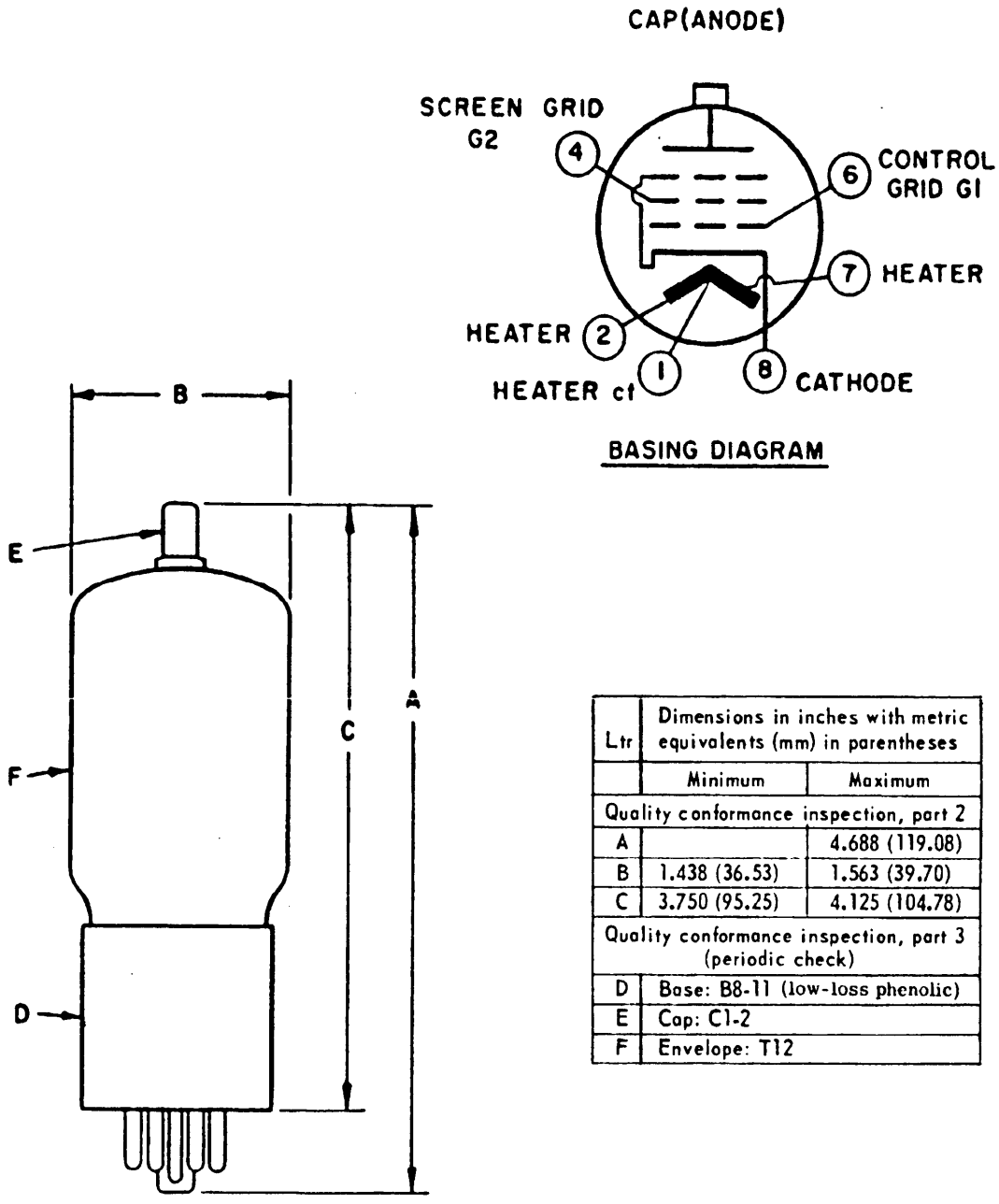
9. All voltages apply simultaneously. There shall be no preheating. The grid current is read after 5 minutes.

Ⓔ 10. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

11. This test shall be performed with an average acceleration level equal to the specified value while being swept through the frequency range of 50 to 500 to 50 Hz. The tubes shall be mounted in each of three planes, X, Y, and Z for one-third of the total excitation time. The frequency in Hz shall be scanned continuously up and down through the entire frequency range at the rate of approximately 2 minutes. The total excitation time for the test shall be 96 hours.

12. The circuit shown on figure 5, or equivalent, shall be used for life-test operation.

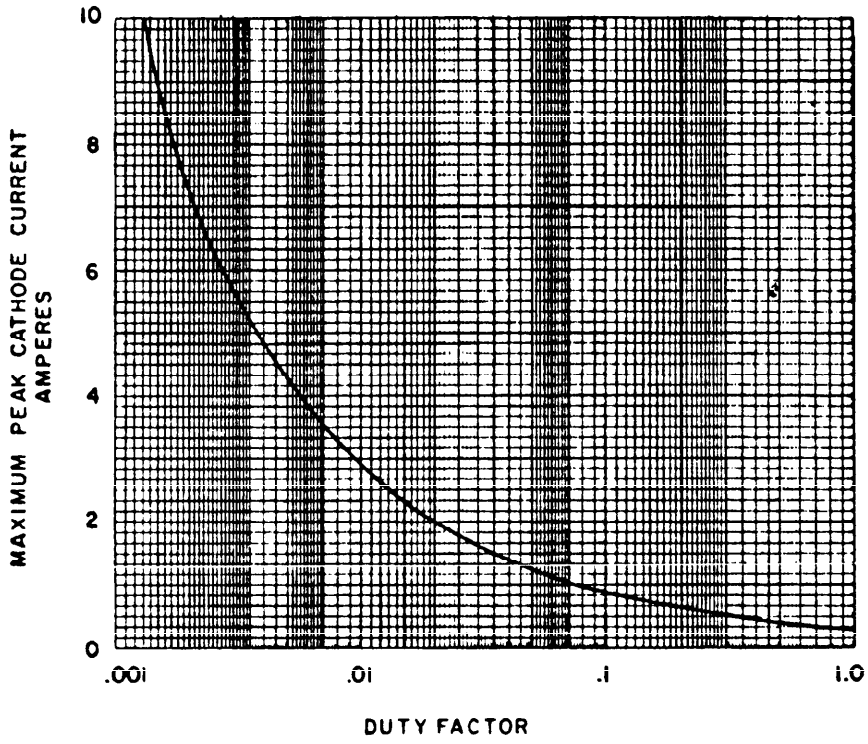
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.688 (119.08)
B	1.438 (36.53)	1.563 (39.70)
C	3.750 (95.25)	4.125 (104.78)
Quality conformance inspection, part 3 (periodic check)		
D	Base: B8-11 (low-loss phenolic)	
E	Cap: C1-2	
F	Envelope: T12	

FIGURE 1. Outline drawing of electron tube type 3D21WB.

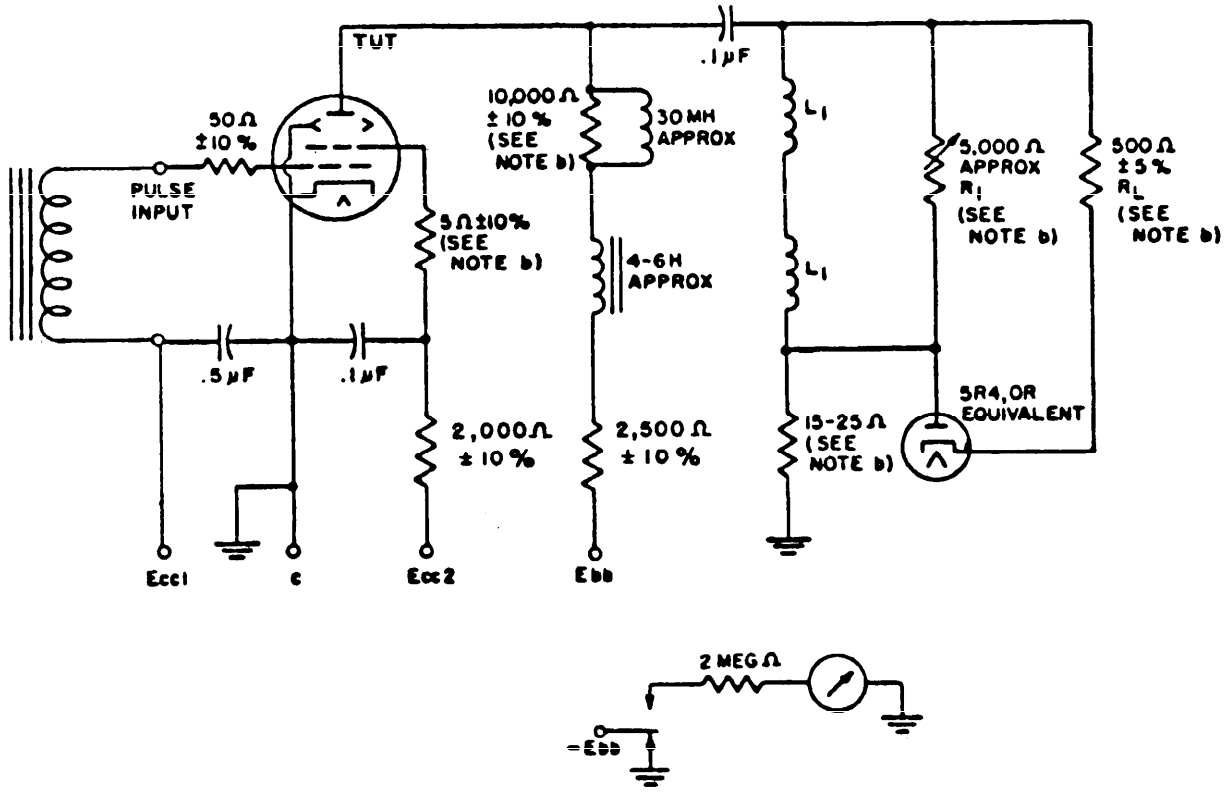
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NOTES:
a. Duty factor (10,000 μs averaging time) is defined as the ratio of 'ON' time in μs to 10,000 μs.
b. 'ON' time is defined as the sum of the duration of all individual pulses which occur during any 10,000-μs interval.

FIGURE 2. Rating chart.

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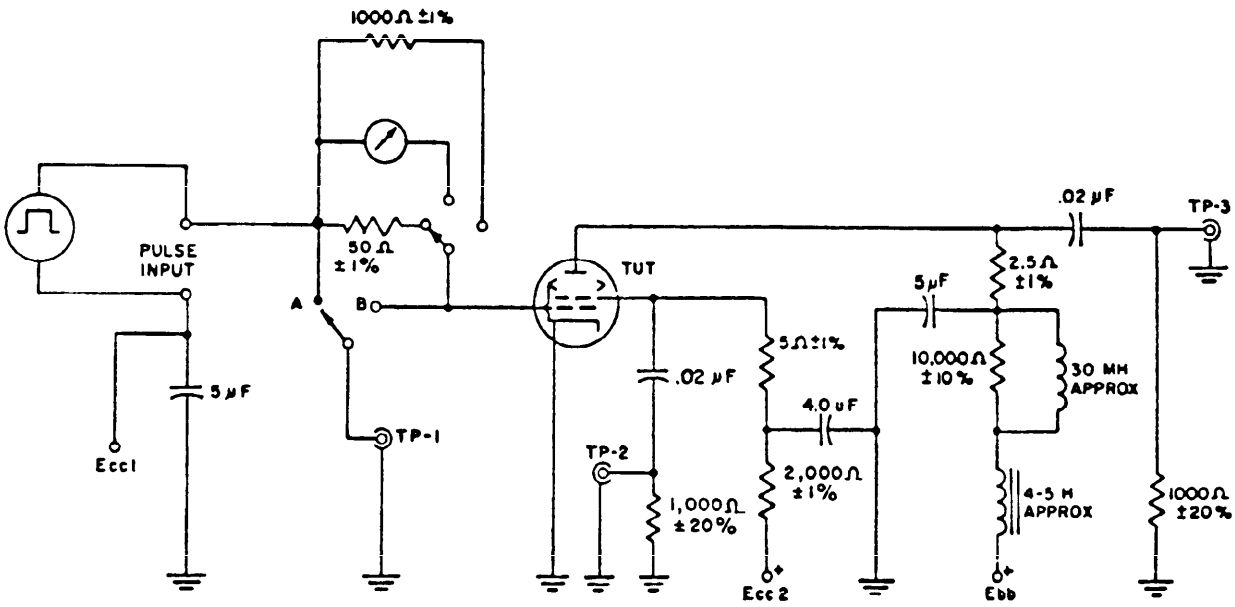


NOTES:

a. L_1 = 5.5 mH approx (Sickles SC101A, or equivalent).
b. Use noninductive resistor.
c. Use paper or mica dielectric capacitors.
d. Anode current is read on a microammeter in series with the negative lead of the power supply and is shown at point E. The 2-megohm resistor and meter are inserted in the negative lead as shown for the anode current test only. L_1 and R_1 adjusted for the voltage backswing = 50 percent of forward swing.

FIGURE 3. Test circuit.

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NOTES:

a. Peak anode current (ib) is read on a calibrated oscilloscope across a 2.5-ohm sensing resistor. (TP-3)

b. Peak screen current (ic_2) is read on a calibrated oscilloscope across a 5.0-ohm sensing resistor. (TP-2)

c. Peak grid current (1) (ic_1) is read on either a calibrated oscilloscope across a 50-ohm sensing resistor ((TP-1) or a milliammeter calibrated for peak reading with duty cycle of 0.001).

d. Peak grid current (2) (ic_1) is read on either a calibrated oscilloscope across a 1,000-ohm sensing resistor (TP-1) or a milliammeter calibrated for peak reading with duty cycle of 0.001.

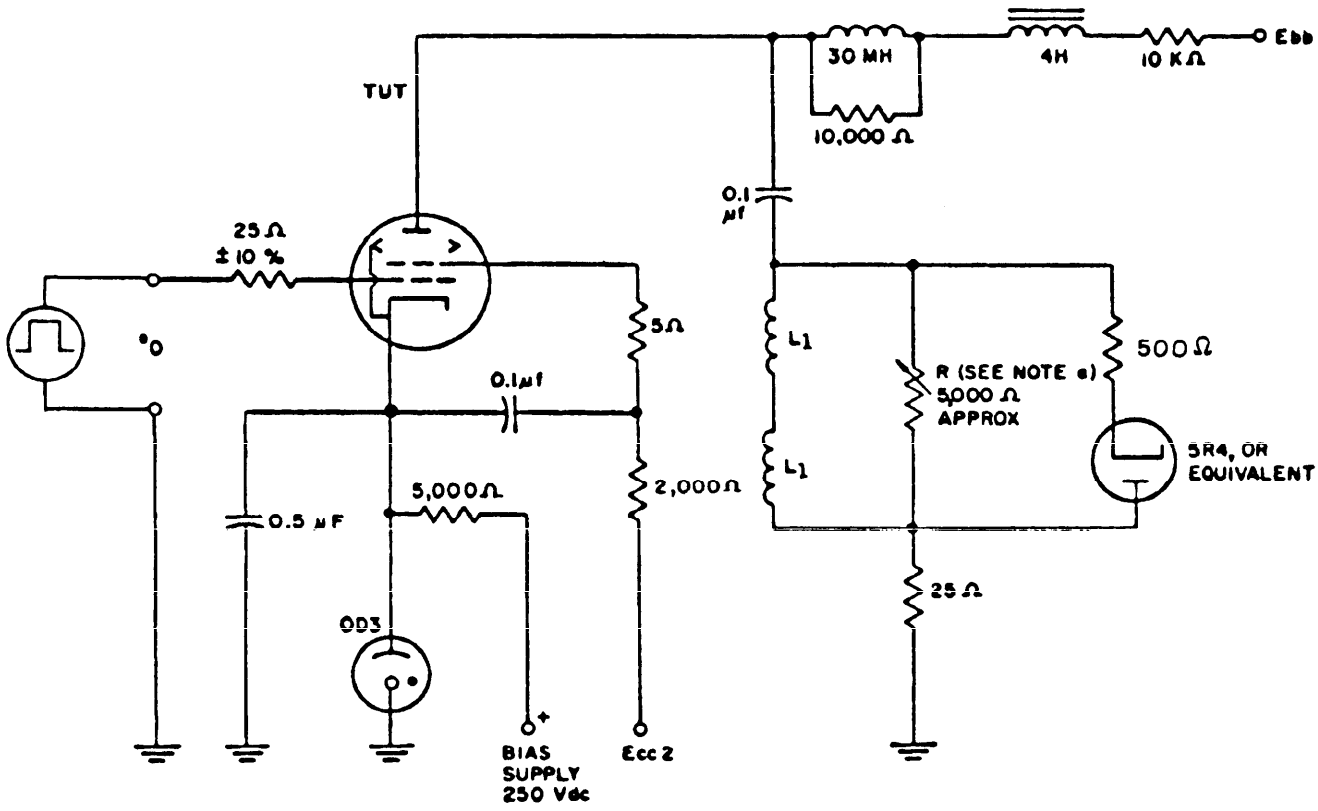
e. egy = e in switch 'B' position. $ic_1(1) = \frac{e \text{ (switch position A)} - e \text{ (switch position B)}}{50}$.
$ic_1(2) = \frac{e \text{ (switch position A)} - e \text{ (switch position B)}}{1000}$.

f. Use noninductive resistors.

g. All capacitors shall be mica or paper dielectric with suitable voltage ratings. TP-1, TP-2, and TP-3 to suitable wide-band oscilloscope with low-capacity probe.

FIGURE 4. Test circuit.

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NOTES:
a. Adjust for 50 percent backswing.
b. L1 = 5.5 mH approx (Sickles SC101A, or equivalent).

FIGURE 5. Test circuit.

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, OS, MC, CG, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (*Mark one*)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (*Specify*):
b. ADDRESS (*Street, City, State, ZIP Code*)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (*Last, First, MI*) – Optional	b. WORK TELEPHONE NUMBER (*Include Area Code*) – Optional
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